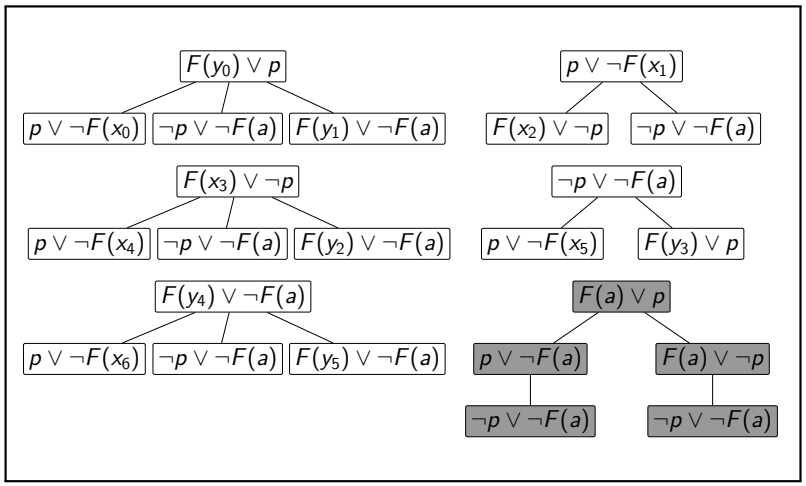
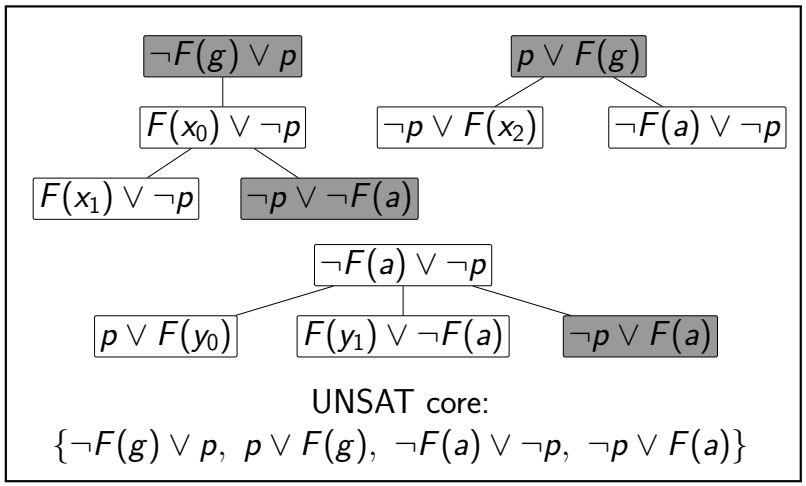
Previous Work

SATCoP improves over LeanCoP by using a SAT solver as unsat proof.

SYN052-1.p

$p \vee \neg F(X)$ $\quad F(X) \vee \neg p$ $\quad F(Y) \vee p$ $\quad \neg p \vee \neg F(a)$ $\quad F(Y) \vee \neg F(a)$





LeanCoP: unsat proof in **43 steps** by closing the Tableau.
The only useful inferences are the ones with the closed tableau.

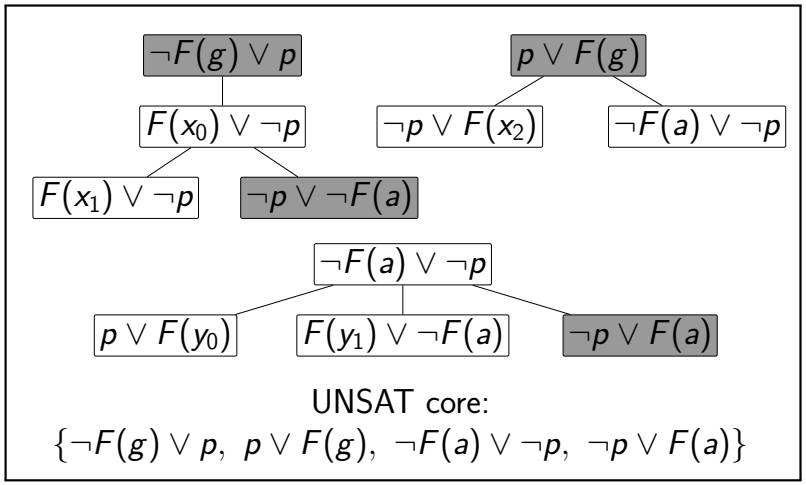
SATCoP: unsat proof in **17 steps** by saturating the SAT core.
The same 4 important steps are found faster.

Michael Rawson and Giles Reger (2021). "Eliminating Models During Model Elimination". In: *Automated Reasoning with Analytic Tableaux and Related Methods*. Ed. by Anupam Das and Sara Negri. Vol. 12842. Cham: Springer International Publishing, pp. 250–265

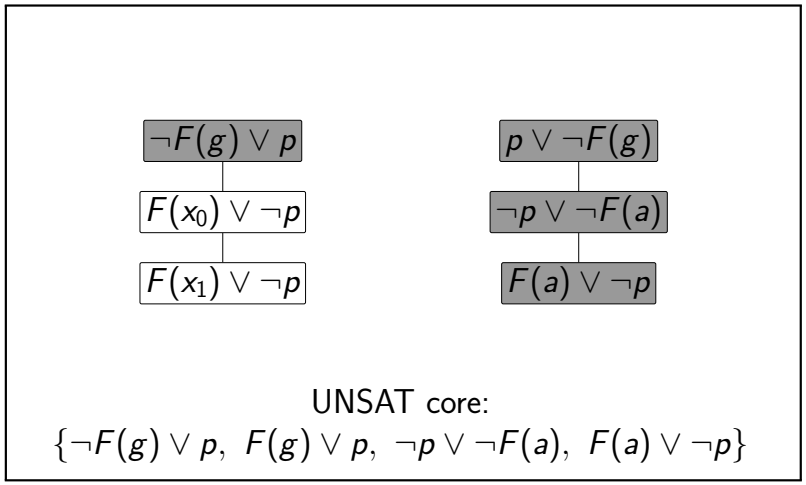
SATResetCoP

Backtracking dominates proof search in connection tableau (Färber 2024)

What if we run Reset every time instead of Backtrack to saturate the UNSAT core faster?



SATCoP: unsat proof in **17 steps** by saturating the SAT core.

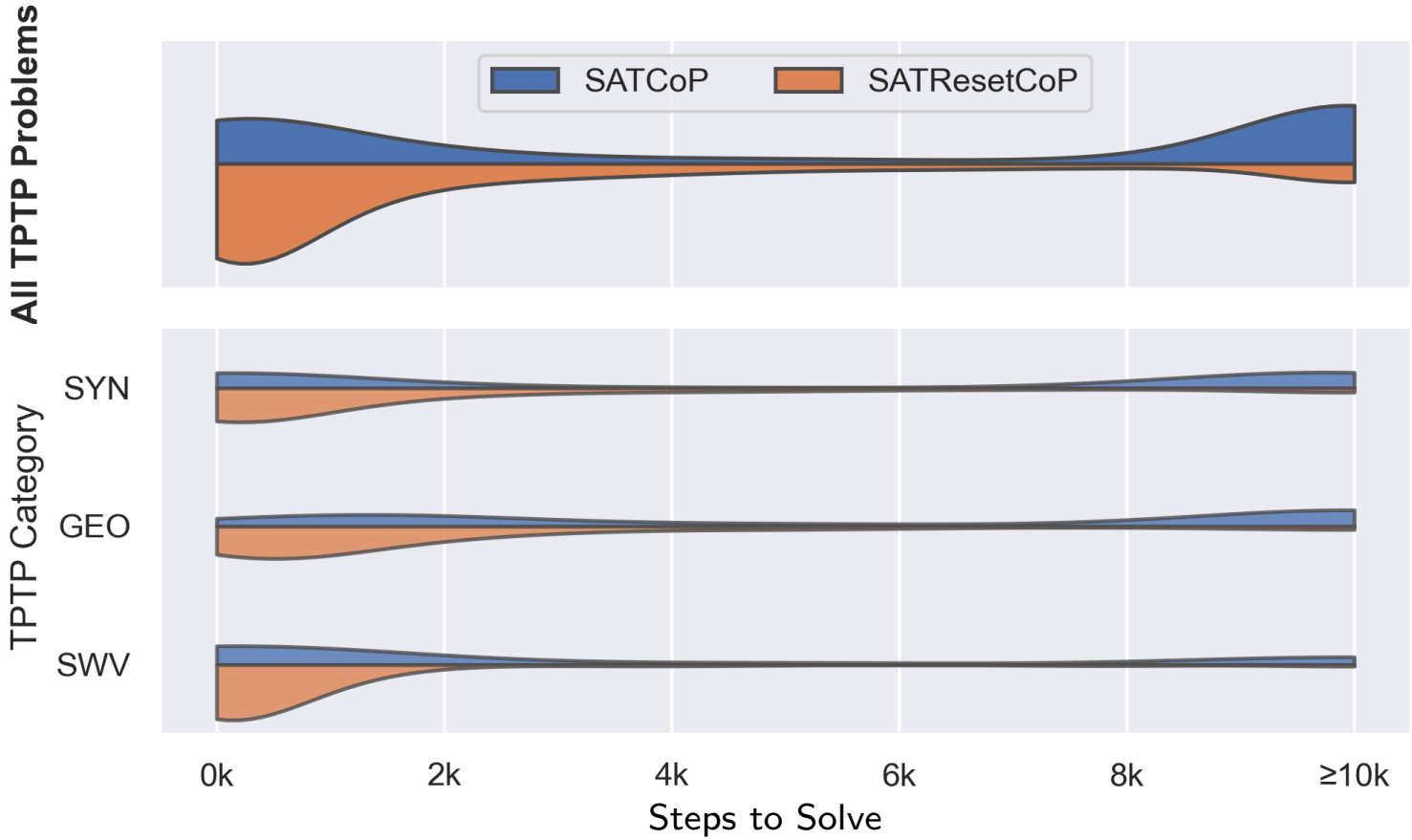


SATResetCoP: unsat proof in **6 steps** by saturating the SAT core. The prover Resets to find the important steps faster.

Michael Färber (Jan. 16, 2024). *A Curiously Effective Backtracking Strategy for Connection Tableaux*. arXiv: 2106.13722 [cs]. URL: http://arxiv.org/abs/2106.13722 (visited on 02/02/2026). Pre-published

SATResetCoP

Good results in almost all TPTP categories!



SATResetCoP improves the SATCoP baseline by 24% when testing with a 90 second time limit.
Almost all problems are solved in fewer steps!